

DSA 8301: Statistical Inference for Big Data

Lecture Notes

Topic 2: Statistical Estimation and Sampling Distributions

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Contents

2	Statistical Estimation and Sampling Distributions	2
2.1	Concept of Point Estimation	2
2.2	Sampling Distributions of Estimators	3
2.2.1	Central Limit Theorem	10
2.3	Estimation Techniques	13
2.3.1	Method of Moments	13
2.3.2	Maximum Likelihood Estimation	18
2.3.3	Least Squares Estimation	23

2 Statistical Estimation and Sampling Distributions

Statistical estimation is a fundamental component of statistical inference. In practice, the true parameters of a population distribution are unknown and must be estimated using information contained in a sample of observed data.

Suppose

$$X_1, X_2, \dots, X_n$$

is a random sample drawn from a population with probability distribution $f(x|\theta)$, where θ is an unknown parameter (or vector of parameters) belonging to a parameter space Θ .

The objective of statistical estimation is to use the observed data to construct estimates of θ .

2.1 Concept of Point Estimation

A **point estimator** is a function of the sample data that provides a single numerical value as an estimate of an unknown parameter.

Definition

Let

$$X_1, X_2, \dots, X_n$$

be a random sample from a distribution depending on parameter θ . A **point estimator** of θ is a statistic

$$\hat{\theta} = T(X_1, X_2, \dots, X_n)$$

that is used to approximate the unknown parameter θ .

Once the sample is observed, the value of the estimator becomes a **point estimate**.

Example

If $X_1, X_2, \dots, X_n$ are independent observations from a population with mean μ , the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is commonly used as a point estimator of μ .

If a sample produces

$$\bar{x} = 12.5$$

then 12.5 is the **point estimate** of μ .

Properties of Point Estimators

Good estimators should possess desirable statistical properties such as:

- Unbiasedness
- Consistency
- Efficiency
- Sufficiency

Details of these properties will be discussed in the next chapter

2.2 Sampling Distributions of Estimators

In statistical inference, unknown population parameters are estimated using statistics computed from a sample. Since these statistics depend on random observations, they themselves are random variables and therefore possess probability distributions.

Understanding these distributions is fundamental because it allows statisticians to measure the variability of estimators and to perform reliable statistical inference.

Definition

Let

$$X_1, X_2, \dots, X_n$$

be a random sample drawn from a population with probability distribution $f(x|\theta)$, where θ is an unknown parameter.

Let

$$\hat{\theta} = T(X_1, X_2, \dots, X_n)$$

be an estimator of θ , where $T(\cdot)$ is a function of the sample.

Definition 2.1. The **sampling distribution** of the estimator $\hat{\theta}$ is the probability distribution of the random variable $\hat{\theta}$ obtained from all possible samples of size n drawn from the population.

In other words, if we repeatedly draw samples of size n from the same population and compute the estimator for each sample, the distribution of these estimator values forms the sampling distribution.

Importance of Sampling Distributions

Sampling distributions are central to statistical inference because they provide information about the variability and reliability of estimators. Specifically, they allow us to:

- quantify estimation uncertainty,

- compute the variance and standard error of estimators,
- construct confidence intervals for unknown parameters,
- perform hypothesis tests,
- compare competing estimators.

Without knowledge of the sampling distribution, it would be impossible to assess how accurate an estimator is.

Standard Error

The variability of an estimator is typically measured using its variance.

$$\text{Var}(\hat{\theta})$$

The square root of this variance is known as the **standard error**:

$$SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}.$$

The standard error quantifies the typical deviation of the estimator from the true parameter value across repeated samples.

Example: Sampling Distribution of the Sample Mean

Consider a random sample

$$X_1, X_2, \dots, X_n$$

drawn independently from a population with mean μ and variance σ^2 .

The sample mean is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Since $\bar{X}$ is a linear combination of random variables, its mean and variance can be derived using properties of expectation.

Mean of the Sample Mean

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}(n\mu) = \mu.$$

Thus, the sample mean is an **unbiased estimator** of the population mean.

Variance of the Sample Mean

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right).$$

Because the observations are independent,

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}.$$

Therefore,

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Exact Sampling Distribution (Normal Case)

If the population distribution is normal,

$$X_i \sim N(\mu, \sigma^2),$$

then the sample mean also follows a normal distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

This result holds for any sample size n .

Central Limit Theorem

In many practical situations the population distribution is unknown or non-normal. A powerful theoretical result known as the **Central Limit Theorem (CLT)** provides an approximation to the sampling distribution of the sample mean.

Theorem 2.1 (Central Limit Theorem). *Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with mean μ and variance $\sigma^2 < \infty$.*

Then as $n \rightarrow \infty$,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Equivalently,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n.$$

This theorem explains why the normal distribution appears so frequently in statistics.

Sampling Distribution of the Sample Variance

Another important estimator is the sample variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

If the underlying population is normally distributed,

$$X_i \sim N(\mu, \sigma^2),$$

then the sampling distribution of the sample variance is

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

This result plays a crucial role in variance estimation and in the construction of confidence intervals for σ^2 .

Illustration 2.2.1

Illustration via Repeated Sampling

Conceptually, the sampling distribution can be understood through the following thought experiment:

1. Draw a random sample of size n from the population.
2. Compute the estimator $\hat{\theta}$.
3. Repeat the process many times.
4. Plot the distribution of the computed estimates.

The resulting distribution approximates the sampling distribution of $\hat{\theta}$ is shown in Figure below.

The same can be demonstrated using Python [CLICK HERE](#)

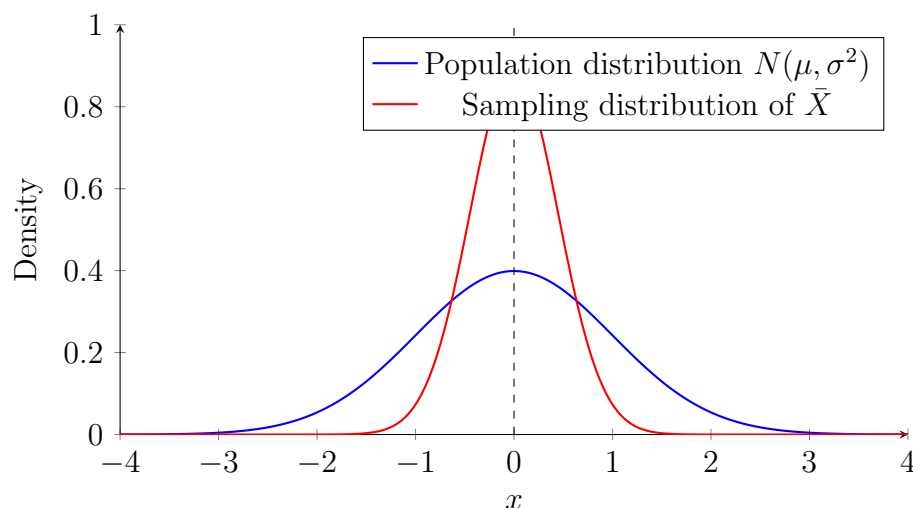


Figure 1: Comparison between a population distribution and the sampling distribution of the sample mean. The sampling distribution has a smaller variance σ^2/n , illustrating that the estimator $\bar{X}$ becomes more concentrated around the true mean μ as the sample size increases.

Example 2.2.1

Sampling Distribution of the Sample Mean

Sampling Distribution of the Sample Mean

Suppose that

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$$

are independent and identically distributed random variables. Thus each observation has mean μ and variance σ^2 :

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

The **sample mean** is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Because $\bar{X}$ is a function of random variables, it is itself a random variable and therefore has its own probability distribution.

Distribution of the Sample Mean

A key property of the normal distribution is that any linear combination of independent normal random variables is also normally distributed.

Since

$$\bar{X} = \frac{1}{n}X_1 + \frac{1}{n}X_2 + \dots + \frac{1}{n}X_n,$$

it is a linear combination of independent normal variables. Therefore

$$\bar{X} \sim N(\mu_{\bar{X}}, \sigma_{\bar{X}}^2)$$

for some mean $\mu_{\bar{X}}$ and variance $\sigma_{\bar{X}}^2$.

We now compute these quantities.

Expectation of the Sample Mean

Using the linearity property of expectation,

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right).$$

Since constants can be factored outside expectations,

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E[X_i].$$

Because each X_i has mean μ ,

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n \mu.$$

Thus

$$E(\bar{X}) = \frac{n\mu}{n} = \mu.$$

Hence the sample mean is an **unbiased estimator** of the population mean.

Variance of the Sample Mean

Next we compute the variance.

$$Var(\bar{X}) = Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right).$$

Using the variance scaling rule,

$$Var(aY) = a^2 Var(Y),$$

we obtain

$$Var(\bar{X}) = \frac{1}{n^2} Var\left(\sum_{i=1}^n X_i\right).$$

Because the observations are independent,

$$Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i).$$

Thus

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2.$$

Since there are n identical terms,

$$\text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2}.$$

Therefore

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Distribution of the Sample Mean

Since $\bar{X}$ is normally distributed with mean μ and variance σ^2/n , we obtain

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

This distribution is known as the **sampling distribution of the sample mean**.

Standard Error

The standard deviation of the sampling distribution is

$$\sqrt{\text{Var}(\bar{X})}.$$

Thus

$$SE(\bar{X}) = \sqrt{\frac{\sigma^2}{n}}.$$

Simplifying,

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

This quantity is called the **standard error of the sample mean**. It measures how much the estimator $\bar{X}$ varies from sample to sample.

Interpretation

Several important conclusions follow:

- The sample mean is centered around the true mean μ .
- The variability of $\bar{X}$ decreases as the sample size n increases.
- Larger samples therefore produce more precise estimates.

In particular, as n increases,

$$\text{Var}(\bar{X}) \rightarrow 0,$$

which explains why the sample mean becomes increasingly concentrated around the true population mean. (*See the figure above*).

2.2.1 Central Limit Theorem

One of the most important results in probability theory and statistical inference is the **Central Limit Theorem (CLT)**. The theorem explains why the normal distribution appears so frequently in statistics and provides the theoretical foundation for many inferential procedures.

In the previous section we established that if the population distribution is normal, then the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is also normally distributed for any sample size n :

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

However, in many practical situations the population distribution is unknown or not normally distributed. The Central Limit Theorem shows that the sampling distribution of the sample mean is approximately normal even when the underlying population distribution is not normal, provided the sample size is sufficiently large.

Formal Statement of the Central Limit Theorem

Let

$$X_1, X_2, \dots, X_n$$

be independent and identically distributed random variables with

$$\mathbb{E}[X_i] = \mu \quad \text{and} \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

Define the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then as the sample size n becomes large,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

The notation $\xrightarrow{d}$ denotes **convergence in distribution**.

Equivalently,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n.$$

Thus, regardless of the original distribution of the data, the distribution of the sample mean becomes approximately normal when the sample size is sufficiently large.

Interpretation

The Central Limit Theorem states that the standardized sample mean

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

behaves approximately like a standard normal random variable when n is large:

$$Z \approx N(0, 1).$$

This result holds for a very broad class of distributions, provided that the observations are independent and the variance of the population is finite.

Why the Central Limit Theorem Is Important

The Central Limit Theorem plays a fundamental role in statistical inference because it allows statisticians to apply normal-based methods even when the population distribution is unknown.

Specifically, it enables us to:

- approximate the sampling distribution of the sample mean
- construct confidence intervals for population parameters
- perform hypothesis tests
- analyze large-sample properties of estimators

Many statistical procedures used in practice rely directly on the CLT.

Effect of Sample Size

The accuracy of the normal approximation improves as the sample size increases.

- If the population distribution is already normal, the sample mean is exactly normal for any n .
- If the population distribution is moderately symmetric, the normal approximation is often good for $n \geq 30$.

- If the population distribution is highly skewed, a larger sample size may be required for the approximation to be accurate. This can be illustrated using Python ([CLICK HERE](#))

Illustration 2.2.2

Graphical Interpretation

Conceptually, the Central Limit Theorem can be illustrated through repeated sampling:

1. Draw many random samples of size n from the population.
2. Compute the sample mean for each sample.
3. Plot the distribution of these sample means.

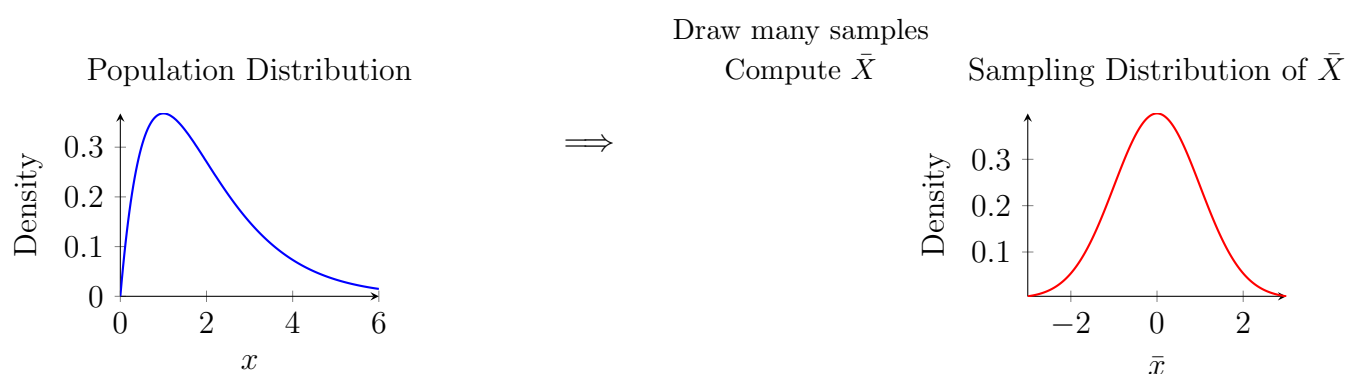


Figure 2: Visual illustration of the Central Limit Theorem. Even if the population distribution is skewed, the distribution of the sample mean becomes approximately normal when the sample size is large.

As the sample size increases, the distribution of the sample means becomes increasingly bell-shaped and approaches the normal distribution.

Example 2.2.2

Suppose a population has mean

$$\mu = 50$$

and standard deviation

$$\sigma = 10.$$

If a sample of size $n = 100$ is drawn, the sampling distribution of the sample mean is approximately

$$\bar{X} \sim N\left(50, \frac{10^2}{100}\right).$$

Thus

$$\bar{X} \sim N(50, 1).$$

The standard error is therefore

$$SE(\bar{X}) = \frac{10}{\sqrt{100}} = 1.$$

This means that the sample mean typically varies around the population mean with a standard deviation of about 1.

2.3 Estimation Techniques

Statistical estimation involves using sample data to infer the values of unknown population parameters. A variety of estimation techniques have been developed, each based on different statistical principles and assumptions.

The choice of estimation method depends on factors such as the underlying statistical model, computational feasibility, and the goals of the analysis.

Common estimation techniques include:

- Method of Moments
- Maximum Likelihood Estimation (MLE)
- Least Squares Estimation
- Bayesian Estimation

Each of these approaches provides a systematic way to construct estimators for unknown parameters.

2.3.1 Method of Moments

The **method of moments (MoM)** is one of the earliest techniques used for parameter estimation in statistics. It was introduced by **Karl Pearson (1894)** and is based on the idea that characteristics of a probability distribution can be summarized using its moments.

The fundamental principle of the method of moments is simple: *estimate unknown parameters by equating theoretical population moments with the corresponding moments computed from the sample.*

Population Moments

Let X be a random variable with distribution depending on a parameter vector

$$\theta = (\theta_1, \theta_2, \dots, \theta_p).$$

The k -th **population moment about the origin** is defined as

$$\mu_k = \mathbb{E}[X^k].$$

These moments are functions of the unknown parameters.

For example:

$$\mu_1 = \mathbb{E}[X], \quad \mu_2 = \mathbb{E}[X^2].$$

Sample Moments

Given a random sample

$$X_1, X_2, \dots, X_n,$$

the k -th sample moment about the origin is

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k.$$

In particular,

$$m_1 = \bar{X}, \quad m_2 = \frac{1}{n} \sum X_i^2.$$

Method of Moments Principle

If the model contains p unknown parameters, the method of moments constructs p equations by equating the first p population moments to the corresponding sample moments:

$$\mu_k(\theta) = m_k, \quad k = 1, \dots, p.$$

Solving these equations yields the method of moments estimators.

Thus the number of moment equations equals the number of unknown parameters.

General Procedure

The method of moments estimation typically follows these steps:

1. Determine the theoretical population moments in terms of the unknown parameters.
2. Compute the corresponding sample moments.
3. Equate population moments to sample moments.
4. Solve the resulting equations for the unknown parameters.

Example 2.3.1

Bernoulli Distribution (Discrete, One Parameter)

Suppose

$$X \sim \text{Bernoulli}(p).$$

The probability mass function is

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

The population mean is

$$\mathbb{E}[X] = p.$$

The first sample moment is

$$m_1 = \bar{X}.$$

Equating the two gives

$$p = \bar{X}.$$

Thus the method of moments estimator is

$$\hat{p}_{MoM} = \bar{X}.$$

Hence the estimated probability of success equals the sample proportion.

Example 2.3.2

Poisson Distribution (Discrete, One Parameter)

Suppose

$$X \sim \text{Poisson}(\lambda).$$

The probability mass function is

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

The population mean is

$$\mathbb{E}[X] = \lambda.$$

Equating the first moment with the sample mean gives

$$\lambda = \bar{X}.$$

Thus

$$\hat{\lambda}_{MoM} = \bar{X}.$$

Example 2.3.3

Exponential Distribution (Continuous, One Parameter)

Suppose

$$X \sim \text{Exponential}(\lambda).$$

The probability density function is

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

The mean of the distribution is

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

The first sample moment is

$$\bar{X}.$$

Equating moments gives

$$\bar{X} = \frac{1}{\lambda}.$$

Solving for λ yields

$$\hat{\lambda}_{MoM} = \frac{1}{\bar{X}}.$$

Example 2.3.4

Example 4: Normal Distribution (Continuous, Two Parameters)

Suppose

$$X \sim N(\mu, \sigma^2).$$

We must estimate two parameters: μ and σ^2 .

The first two population moments are

$$\mathbb{E}[X] = \mu$$

$$\mathbb{E}[X^2] = \mu^2 + \sigma^2.$$

The corresponding sample moments are

$$m_1 = \bar{X}$$

$$m_2 = \frac{1}{n} \sum X_i^2.$$

Equating moments gives

$$\mu = \bar{X}$$

$$\mu^2 + \sigma^2 = m_2.$$

Substituting $\hat{\mu} = \bar{X}$ yields

$$\hat{\sigma}_{MoM}^2 = m_2 - \bar{X}^2.$$

Thus the MoM estimators are

$$\hat{\mu}_{MoM} = \bar{X},$$

$$\hat{\sigma}_{MoM}^2 = \frac{1}{n} \sum X_i^2 - \bar{X}^2.$$

Example 2.3.5

Example 5: Gamma Distribution (Continuous, Two Parameters)

Suppose

$$X \sim \text{Gamma}(\alpha, \beta)$$

with

$$\mathbb{E}[X] = \alpha\beta,$$

$$\text{Var}(X) = \alpha\beta^2.$$

Using the relationships

$$\bar{X} \approx \alpha\beta$$

$$S^2 \approx \alpha\beta^2,$$

we obtain

$$\hat{\beta} = \frac{S^2}{\bar{X}},$$

$$\hat{\alpha} = \frac{\bar{X}^2}{S^2}.$$

These are the method of moments estimators.

Advantages of the Method of Moments

- Conceptually simple and easy to apply.
- Often produces explicit closed-form estimators.

- Requires minimal distributional assumptions.
- Useful when maximum likelihood estimation is difficult.

Limitations of the Method of Moments

- Estimators may not be statistically efficient.
- They may lie outside the parameter space.
- MoM estimators may have higher variance compared to MLE.
- They do not always use all available information in the likelihood.

Conclusion

The method of moments provides a straightforward technique for parameter estimation by matching theoretical and empirical moments. Although it is often less efficient than maximum likelihood estimation, it remains an important tool in statistical modeling, particularly when likelihood-based methods are difficult to apply.

2.3.2 Maximum Likelihood Estimation

The **maximum likelihood estimation (MLE)** method is one of the most widely used estimation techniques in statistics.

The basic idea is to choose parameter values that maximize the likelihood of observing the data.

Suppose

$$X_1, X_2, \dots, X_n \sim f(x|\theta)$$

are independent observations from a distribution with parameter θ .

The likelihood function is

$$L(\theta) = \prod_{i=1}^n f(x_i|\theta).$$

The maximum likelihood estimator is defined as

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} L(\theta).$$

Illustration 2.3.1

Meaning and Pronunciation of $\arg \max$

1. How to Pronounce It The expression

$$\arg \max_{\theta \in \Theta} L(\theta)$$

is usually read as:

“the argument that maximizes $L(\theta)$ over θ in Θ .”

2. Conceptual Meaning Each part of the expression has a specific interpretation:

- $\arg$: argument (the input value of the function)
- $\max$: maximum
- $L(\theta)$: a function we want to maximize (often a likelihood function)
- θ : a parameter
- Θ : the parameter space

Thus, the expression

$$\arg \max_{\theta \in \Theta} L(\theta)$$

means:

the value of θ that makes $L(\theta)$ as large as possible.

3. Mathematical Meaning In statistical estimation we often write

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta).$$

This means that

$$\hat{\theta}$$

is the value of θ that maximizes the function $L(\theta)$.

In the context of maximum likelihood estimation (MLE), this value is called the **maximum likelihood estimator**.

4. Difference Between max and arg max It is important to distinguish between these two related concepts.

Maximum Value

$$\max_{\theta \in \Theta} L(\theta)$$

gives the **largest value of the function**.

Argument of the Maximum

$$\arg \max_{\theta \in \Theta} L(\theta)$$

gives the **value of θ at which the maximum occurs**.

Often we work with the log-likelihood

$$\ell(\theta) = \log L(\theta),$$

because logarithms transform products into sums.

Example 2.3.6

Example 1: Bernoulli Distribution (Discrete, One Parameter) Suppose

$$X_1, \dots, X_n \sim \text{Bernoulli}(p).$$

The probability mass function is

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

The likelihood function is

$$L(p) = \prod_{i=1}^n p^{x_i}(1 - p)^{1-x_i}.$$

This simplifies to

$$L(p) = p^{\sum x_i}(1 - p)^{n - \sum x_i}.$$

Taking logarithms gives

$$\ell(p) = \sum x_i \log p + (n - \sum x_i) \log(1 - p).$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p}.$$

Setting equal to zero yields

$$\hat{p}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}.$$

Thus the MLE of p is the sample proportion.

Example 2: Poisson Distribution (Discrete, One Parameter) Suppose

$$X_1, \dots, X_n \sim \text{Poisson}(\lambda).$$

The pmf is

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

The likelihood is

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}.$$

Simplifying,

$$L(\lambda) = \lambda^{\sum x_i} e^{-n\lambda} \prod \frac{1}{x_i!}.$$

The log-likelihood is

$$\ell(\lambda) = \left(\sum x_i \right) \log \lambda - n\lambda + C.$$

Differentiating,

$$\frac{d\ell}{d\lambda} = \frac{\sum x_i}{\lambda} - n.$$

Solving,

$$\hat{\lambda}_{MLE} = \bar{X}.$$

Example 3: Exponential Distribution (Continuous, One Parameter)
Suppose

$$X_1, \dots, X_n \sim \text{Exponential}(\lambda).$$

The density is

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i}.$$

Simplifying,

$$L(\lambda) = \lambda^n e^{-\lambda \sum x_i}.$$

Log-likelihood:

$$\ell(\lambda) = n \log \lambda - \lambda \sum x_i.$$

Differentiating,

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum x_i.$$

Thus

$$\hat{\lambda}_{MLE} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}.$$

Example 4: Normal Distribution (Continuous, One Parameter) Suppose

$$X_i \sim N(\mu, \sigma^2)$$

with σ^2 known.

The likelihood function is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Taking logs,

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 + C.$$

Differentiating and solving gives

$$\hat{\mu}_{MLE} = \bar{X}.$$

Example 5: Normal Distribution (Two Parameters) Suppose

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

with both μ and σ^2 unknown.

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2.$$

Differentiating with respect to μ ,

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) = 0.$$

Thus

$$\hat{\mu} = \bar{X}.$$

Next differentiate with respect to σ^2 ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2.$$

Solving,

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum (x_i - \bar{X})^2.$$

Example 6: Gamma Distribution (Two Parameters) Suppose

$$X_1, \dots, X_n \sim \text{Gamma}(\alpha, \beta).$$

The density is

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}.$$

The likelihood becomes

$$L(\alpha, \beta) = \prod \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} e^{-x_i/\beta}.$$

The log-likelihood is

$$\ell(\alpha, \beta) = -n \log \Gamma(\alpha) - n\alpha \log \beta + (\alpha - 1) \sum \log x_i - \frac{1}{\beta} \sum x_i.$$

The likelihood equations must be solved numerically.

Properties of MLE Under regularity conditions, MLEs have several desirable properties:

- Consistency
- Asymptotic normality
- Asymptotic efficiency

2.3.3 Least Squares Estimation

The **least squares method** is one of the most widely used techniques for estimating unknown parameters in statistical models, particularly in regression analysis. The method was originally developed by **Carl Friedrich Gauss** and **Adrien-Marie Legendre** in the early 19th century.

The central idea of least squares estimation is to determine parameter values that minimize the discrepancy between the observed data and the values predicted by the model.

Simple Linear Regression Model

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where

- Y_i is the response variable,
- X_i is the explanatory variable,
- β_0 is the intercept,
- β_1 is the slope parameter,
- ε_i represents random error terms.

The errors are typically assumed to satisfy

$$E(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

Thus the conditional mean of Y_i is

$$E(Y_i|X_i) = \beta_0 + \beta_1 X_i.$$

Residuals

For each observation, the difference between the observed value and the predicted value is called the **residual**:

$$e_i = Y_i - \hat{Y}_i.$$

Since

$$\hat{Y}_i = \beta_0 + \beta_1 X_i,$$

the residual becomes

$$e_i = Y_i - (\beta_0 + \beta_1 X_i).$$

Least Squares Criterion

The least squares method estimates β_0 and β_1 by minimizing the **sum of squared residuals**:

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Squaring ensures that positive and negative errors do not cancel each other and also penalizes larger errors more heavily.

Derivation of the Least Squares Estimators

To find the values of β_0 and β_1 that minimize S , we take partial derivatives with respect to the parameters and set them equal to zero.

$$\frac{\partial S}{\partial \beta_0} = -2 \sum (Y_i - \beta_0 - \beta_1 X_i) = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum X_i (Y_i - \beta_0 - \beta_1 X_i) = 0$$

These equations are known as the **normal equations**.

Solving them simultaneously yields the least squares estimators.

Estimator of the Slope

The estimator of the slope parameter is

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}.$$

This estimator measures the strength and direction of the linear relationship between X and Y .

Estimator of the Intercept

The intercept estimator is

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Thus the fitted regression line becomes

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X.$$

Interpretation of the Coefficients

- $\hat{\beta}_1$ represents the expected change in Y for a one-unit increase in X .
- $\hat{\beta}_0$ represents the expected value of Y when $X = 0$.

Geometric Interpretation

The least squares solution can also be interpreted geometrically. The fitted regression line minimizes the vertical distances between the observed data points and the predicted values.

Thus, the method finds the line that provides the **best linear fit** to the data in terms of minimizing squared errors.

Matrix Formulation

In matrix notation, the linear regression model can be written as

$$\mathbf{Y} = X\beta + \varepsilon,$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

The least squares estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

This formula generalizes easily to multiple regression models with several explanatory variables.

Properties of Least Squares Estimators

Under the classical linear regression assumptions, the least squares estimators have several desirable properties:

- They are **unbiased**.
- They have **minimum variance** among all linear unbiased estimators (Gauss–Markov theorem).
- They are **consistent**.

Connection with Maximum Likelihood Estimation

If the regression errors follow a normal distribution

$$\varepsilon_i \sim N(0, \sigma^2),$$

then the likelihood function for the observations is

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2}\right).$$

Taking logarithms gives

$$\ell(\beta_0, \beta_1) = -\frac{1}{2\sigma^2} \sum (Y_i - \beta_0 - \beta_1 X_i)^2 + C.$$

Maximizing this likelihood is equivalent to minimizing

$$\sum (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Therefore,

$$\hat{\beta}_{LS} = \hat{\beta}_{MLE}.$$

Thus the least squares estimators coincide with the maximum likelihood estimators when the regression errors are normally distributed.

Try Your Knowledge 2.1

Practical Exercises

1. Point Estimation

Let $X_1, X_2, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 .

- (a) Define a point estimator.

- (b) Show that the sample mean $\bar{X}$ is an unbiased estimator of μ .
- (c) If a sample of size $n = 8$ yields values

4, 6, 5, 7, 3, 5, 6, 4,

compute the point estimate of μ .

2. Sampling Distribution of the Sample Mean

Suppose

$$X_1, X_2, \dots, X_{25} \sim N(50, 16).$$

- (a) Determine the sampling distribution of $\bar{X}$.
- (b) Compute the mean and variance of $\bar{X}$.
- (c) Compute the standard error of $\bar{X}$.

3. Standard Error

A population has standard deviation $\sigma = 20$.

- (a) Compute the standard error of the sample mean when $n = 25$.
- (b) Compute the standard error when $n = 100$.
- (c) Explain how sample size affects the precision of an estimator.

4. Central Limit Theorem

A population has mean $\mu = 40$ and standard deviation $\sigma = 12$.

- (a) For a sample size $n = 36$, determine the approximate distribution of $\bar{X}$.
- (b) Compute the standard error.
- (c) Find the probability that $\bar{X} > 43$ using the CLT.

5. Sampling Distribution of Sample Variance

Suppose

$$X_1, \dots, X_{15} \sim N(\mu, \sigma^2).$$

- (a) State the sampling distribution of the sample variance S^2 .
- (b) Identify the degrees of freedom.
- (c) Write the distribution of

$$\frac{(n-1)S^2}{\sigma^2}.$$

6. Method of Moments: Bernoulli Distribution

Suppose

$$X_1, \dots, X_n \sim \text{Bernoulli}(p).$$

- (a) Derive the method of moments estimator of p .
- (b) If a sample of size 20 contains 12 successes, estimate p .

7. Method of Moments: Poisson Distribution

Suppose

$$X_1, \dots, X_n \sim \text{Poisson}(\lambda).$$

- (a) Derive the method of moments estimator of λ .
- (b) If the sample mean is 3.8, estimate λ .

8. Method of Moments: Exponential Distribution

Suppose

$$X_1, \dots, X_n \sim \text{Exponential}(\lambda).$$

- (a) Derive the method of moments estimator of λ .
- (b) If the sample mean is 5, compute the estimate.

9. Method of Moments: Normal Distribution

Suppose

$$X_1, \dots, X_n \sim N(\mu, \sigma^2).$$

- (a) Derive the method of moments estimators for μ and σ^2 .
- (b) Explain why two moment equations are required.

10. Maximum Likelihood Estimation: Bernoulli

Let

$$X_1, \dots, X_n \sim \text{Bernoulli}(p).$$

- (a) Write the likelihood function.
- (b) Derive the log-likelihood function.
- (c) Obtain the MLE of p .

11. Maximum Likelihood Estimation: Poisson

Suppose

$$X_1, \dots, X_n \sim \text{Poisson}(\lambda).$$

- (a) Write the likelihood function.
- (b) Derive the log-likelihood.
- (c) Obtain the maximum likelihood estimator of λ .

12. Maximum Likelihood Estimation: Exponential

Suppose

$$X_1, \dots, X_n \sim \text{Exponential}(\lambda).$$

- (a) Write the likelihood function.
- (b) Derive the log-likelihood function.
- (c) Obtain the MLE of λ .

13. MLE for Normal Distribution

Suppose

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

with σ^2 known.

- (a) Write the likelihood function.
- (b) Derive the log-likelihood.
- (c) Obtain the maximum likelihood estimator of μ .

14. Least Squares Estimation

Consider the regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

- (a) Define the least squares criterion.
- (b) Derive the normal equations.
- (c) Write the formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$.

15. Regression Calculation

Suppose the following data are observed:

X	1	2	3	4	5
Y	2	3	5	4	6

- (a) Compute $\bar{X}$ and $\bar{Y}$.
- (b) Compute the least squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$.
- (c) Write the fitted regression equation.